Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

On changing N_2 to N_2^+, the dissociation energy of $N-N$ bond . . . . . and on changing O_2 to O_2^+ the dissociation energy of O-O bond....

(A) increases, decreases

(B) decreases, increases

(C) decreases in both cases

(D) increases in both cases

Q.2

Which of the following is correct increasing order of lone pair of electrons on the central atom?

(A) $IF_7 < IF_5 < ClF_3 < XeF_2$

(B) $IF_7 < XeF_2 < ClF_2 < IF_5$

(C) $IF_7 < ClF_3 < XeF_2 < IF_5$

(D) $IF_7 < XeF_2 < IF_5 < ClF_3$

Q.3

N_2 and O_2 are converted into monocations, N_2^+ and O_2^+ respectively. Which of the following statements is wrong?

(A) In N_2^+, $N-N$ bond weakens

(B) In O_2^+, the $O-O$ bond order increases

(C) In O_2^+, paramagnetism decreases

(D) N_2^+ becomes diamagnetic

Q.4

$Cl-O$ bond order in perchlorate ion is ___

(A) 1.75

(B) 1.85

(C) 2.75

(D) 1.80

@IITJEE_Advanced

Q.5

The bond dissociation energy of $B-F$ in BF_3 is $646\ kJ\ mol^{-1}$ whereas that of $C-F$ in CF_4 is 515 $kJmol^{-1}$. The correct reason for higher $B-F$ bond dissociation energy as compared to that of $C-F$ is

(A) stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4

(B) significant $p\pi-p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4

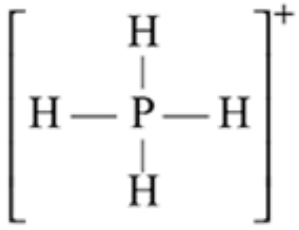
(C) lower degree of $p\pi-p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4.

(D) Smaller size of B-atom as compared to that of C -atom.

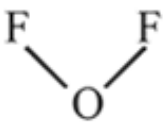
Q.6

Which one of the following formulae does not correctly represent the bonding capacities of the two atoms involved?

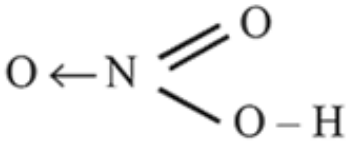
(A)



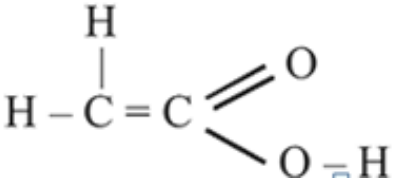
(B)



(C)



(D)



Q.7

Arrange the following ions in the order of decreasing $X-O$ bond length, where X is the central atom

(A) $ClO_4^-, SO_4^{2-}, PO_4^{3-}, SiO_4^-$

(B) $SiO_4^{4-}, PO_4^{3-}, SO_4^{2-}, ClO_4^-$

(C) $SiO_4^{4-}, PO_4^{3-}, ClO_4^-, SO_4^{2-}$

(D) $SiO_4^{4-}, SO_4^{2-}, PO_4^{3-}, ClO_4^-$

@IITJEE_Advanced

Q.8

In which of the following ionization processes the bond energy has increased and also the magnetic behaviour has changed from paramagnetic to diamagnetic?

(A) $\mathrm{NO} \rightarrow \mathrm{NO}^{+}$

(B) $\mathrm{N}_2 \rightarrow \mathrm{N}_2^{+}$

(C) $\mathrm{C}_2 \rightarrow \mathrm{C}_2^{+}$

(D) $\mathrm{O}_2 \rightarrow \mathrm{O}_2^{+}$

Q.9

The electronic configuration of four elements $\mathrm{L}, \mathrm{P}, \mathrm{Q}$ and R are given in brackets
$\mathrm{L}\left(1\ \mathrm{s}^2, 2\ \mathrm{s}^2 2\mathrm{p}^4\right) ; \mathrm{Q}\left(\mathrm{ls}^2, 2\ \mathrm{s}^2 2\mathrm{p}^6, 3\ \mathrm{s}^2 3\mathrm{p}^5\right)$
$\mathrm{P}\left(\mathrm{ls}^2, 2\ \mathrm{s}^2 2\mathrm{p}^6, 3\ \mathrm{s}^1\right) ; \mathrm{R}\left(\mathrm{ls}^2, 2\ \mathrm{s}^2 2\mathrm{p}^6, 3\ \mathrm{s}^2\right)$
The formulae of ionic compounds that can be formed between these elements are

(A) L_2P, RL, PQ and R_2Q

(B) LP, RL, PQ and RQ

(C) P_2L, RL, PQ and RQ_2

(D) LP, R_2L, P_2Q and RQ

Q.10

Match List I (species) and List II (bond orders) and select the correct answer:

List-I	List-II
A. N_2	1.1.0
B. O_2	2.2.0
C. F_2	3.2.5
D. O_2^{+}	4. 3.0

(A) $\mathrm{A} \rightarrow 4,\ \mathrm{B} \rightarrow 2, \mathrm{C} \rightarrow 1, \mathrm{D} \rightarrow 3$

(B) $\mathrm{A} \rightarrow 1,\ \mathrm{B} \rightarrow 3, \mathrm{C} \rightarrow 4, \mathrm{D} \rightarrow 2$

(C) $\mathrm{A} \rightarrow 1,\ \mathrm{B} \rightarrow 2, \mathrm{C} \rightarrow 4, \mathrm{D} \rightarrow 3$

(D) $\mathrm{A} \rightarrow 4,\ \mathrm{B} \rightarrow 3, \mathrm{C} \rightarrow 1, \mathrm{D} \rightarrow 2$

Q.11

The bond order of CN^{-} is

(A) 2

(B) 2.5

(C) 3

@IITJEE_Advanced

(D) 1.5

Q.12

Which one of the following species will have maximum bond dissociation energy?

(A) C_2

(B) C_2^+

(C) C_2^-

(D) C_2^{2-}

Q.13

Which of the following statements regarding valence-bond method is not true?

(A) In the valence-bond method, the molecule is considered to be the collection of atoms and then interactions between different atoms are considered

(B) For a molecule to be stable, the electrostatic attractions must predominate over the electrostatic repulsions

(C) The potential energy of a diatomic molecule is less than the sum of corresponding energies of free atoms

(D) The net force of attraction acting on the atoms in a molecule is greater than that of repulsion

Q.14

Which of the following species is isoelectronic with CO?

(A) N_2^+

(B) CN^-

(C) O_2^+

(D) O_2^-

Q.15

The geometry of ClO_3^- ion according to Valence Shell Electron Pair Repulsion (VSEPR) theory will be:

(A) Planar triangular

(B) Pyramidal

(C) Tetrahedral

(D) Square planar

Answers & Solutions

Q.1 Answer:

@IITJEE_Advanced

decreases, increases

Solution:

On changing N_2 to N_2^+, B.O. decreases from 3 to 2.5 whereas on changing O_2 to O_2^+, B.O. increases from 2 to 2.5. In former case, the bond dissociation energy decreases and in the latter case, it increases.

Q.2 Answer:

$IF_7 < IF_5 < ClF_3 < XeF_2$

Solution:

The number of lone pairs of electrons on central atom in various given species are

\begin{array}{cc}

**Species & Number of lone pairs on central atom **

**\hline \mathrm{IF}_{7} & 0 **

**\hline \mathrm{IF}_{5} & 1 **

**\hline \mathrm{ClF}_{3} & 2 **

**\hline \mathrm{XeF}_{2} & 3 **

\hline

\end{array}

Thus the correct increasing order is

$IF_7 < IF_5 < ClF_3 < XeF_2$

Q.3 Answer:

N_2^+ becomes diamagnetic

Solution:

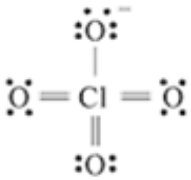
$\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \left(\pi_{2p}^2 = \pi_{2p}^2\right) \sigma_{2p}^1$ ($N_2^+ = 13$ **electrons**)

it contains one unpaired electron hence paramagnetic.

Q.4 Answer:

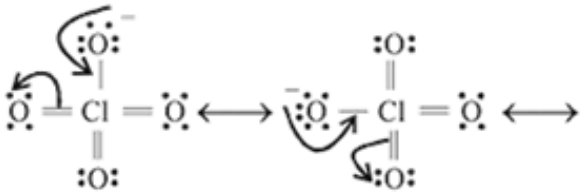
1.75

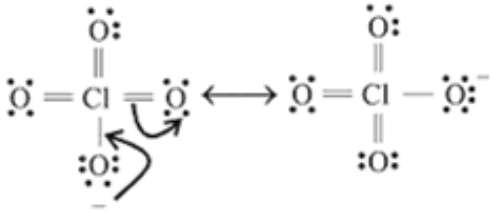
Solution:



@IITJEE_Advanced

Perchlorate ion $\left(ClO_4^-\right)$ have the following possible resonating structures:





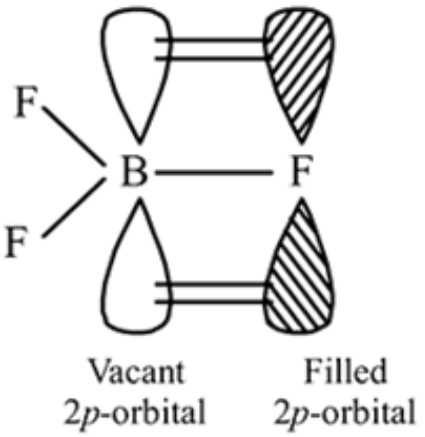
$Cl-O$ bond order $= 7/4 = 1.75$

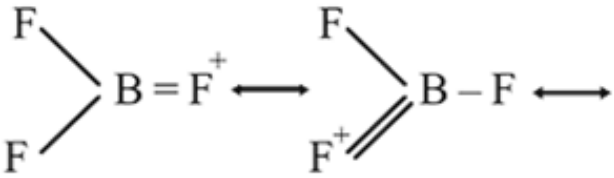
Q.5 Answer:

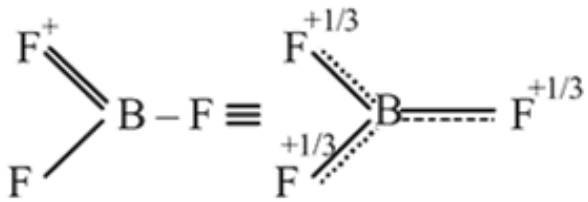
significant $p\pi - p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4

Solution:

The delocalised $p\pi - p\pi$ bonding between filled p -orbital of F and vacant p -orbital of B leads to shortening of B-F bond length which results in higher bond dissociation energy of the B-F bond.

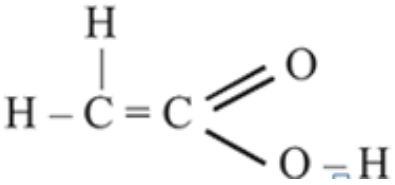




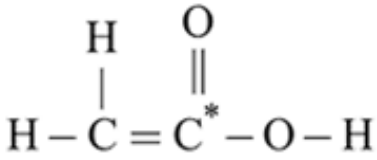


Q.6 Answer:

@IITJEE_Advanced



Solution:



The star marked carbon has a valency of 5 and hence this formula is not correct.

Q.7 Answer:

$\mathrm{SiO_4^{4-}}, \mathrm{PO_4^{3-}}, \mathrm{SO_4^{2-}}, \mathrm{ClO_4^-}$

Solution:

More will be the electronegativity of X, lesser will be the bond length of X -O bond.

Q.8 Answer:

$\mathrm{NO} \rightarrow \mathrm{NO^+}$

Solution:

For NO

Total no. of electrons $= 15$

$\mathrm{B \cdot O} = 2.5$

Mag. Behaviour = Paramagnetic

For $\mathrm{NO^+}$

Total no. of electrons $= 14$

B. $\mathrm{O} = 3$

Mag. Behaviour $=$ **Diamagnetic**

Q.9 Answer:

P_2L, RL, PQ and RQ_2

Solution:

Valencies of L, Q, P and R is $-2, -1, +1$ and $+2$ respectively so they will form P_2L, RL, PQ and RQ_2

Q.10 Answer:

$\mathrm{A} \rightarrow 4,\ \mathrm{B} \rightarrow 2, \mathrm{C} \rightarrow 1, \mathrm{D} \rightarrow 3$

Solution:

@IITJEE_Advanced

$N_2 \rightarrow \sigma(2s)^2\sigma(2s)^{*2}\pi(2Px)^2\pi(2py)^2\sigma(2pz)^2$ **Bond order** $= \frac{1}{2}(8-2) = 3$
$O_2 \rightarrow$ **bond order** $= \frac{1}{2}(8-4) = 2$
$F_2 \rightarrow$ **bond order** $= \frac{1}{2}(8-6) = 1$
$O_2^+ \rightarrow$ **bond order** $= \frac{1}{2}(8-3) = 2.5$

Q.11 Answer:

3

Solution:

Valence electrons in $\mathrm{CN}^- = 10$ **Electronic configuration** $(\sigma 2\,\mathrm{s})^2(\sigma * 2\,\mathrm{s})^2\,(\pi 2\mathrm{p}_x)^2\,(\pi 2\mathrm{p}_y)^2\,(\sigma 2\mathrm{p}_z)^2$
Bond order $= \frac{\text{(Bonding - antibonding) electrons}}{2} = \frac{8-2}{2} = 3$

Q.12 Answer:

C_2^{2-}

Solution:

C_2**(8 electrons);** $\mathrm{KK}(\sigma 2\,\mathrm{s})^2(\sigma * 2\,\mathrm{s})^2\,(\pi * 2\mathrm{p}_x)^2\,(\pi 2\mathrm{p}_y)^2$ **Bond order** $= (6-2)/2 = 2$
C_2^+**(7 electrons); Bond order** $= (5-2)/2 = 11/2$
C_2^-**(9 electrons);** $\mathrm{KK}(\sigma 2\,\mathrm{s})^2(\sigma * 2\,\mathrm{s})^2\,(\pi 2\mathrm{p}_x)^2\,(\pi 2\mathrm{p}_y)^2\,(\sigma 2\mathrm{p}_y)^1$ **Bond order** $= (7-2)/2 = 2\frac{1}{2}$
C_2^{2-}**(10 electrons); Bond order** $= (8-2)/2 = 3$
Larger the bond order, larger the dissociation energy.

Q.13 Answer:

The net force of attraction acting on the atoms in a molecule is greater than that of repulsion

Solution:

The net force of attraction acting on the atom in a molecule is greater than that of repulsion.

Q.14 Answer:

CN^-

Solution:

CO **and** CN^- **are isoelectronic.**

$CO = 6 + 8 = 14$ **and** $CN^- = 6 + 7 + 1 = 14$ **.**

Q.15 Answer:

Pyramidal

Solution:

@IITJEE_Advanced

Cl has 3 bond pairs and 1 lone pair in ClO_3^-. So, it is pyramidal.

@IITJEE_Advanced